AN IRREDUCIBLE SCHEME WITH A DENSE SET OF RATIONAL POINTS IS GEOMETRICALLY IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the contradiction; no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-irreducible-dense-rational-points-geometrically-irreducible is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Characterization of geometric irreducibility by finite separable field extensions. Same chapter, Lemma lemma-characterize-geometrically-irreducible, invoked at proof lines 1681-1682. Supplies that it is enough to show irreducibility of the base change along finite extensions.
- Irreducible components of a base change project onto irreducible components. Same chapter, Lemma lemma-image-irreducible, invoked at proof lines 1683-1684. Supplies that both chosen components project onto X.
- Image of a nowhere dense subset under a finite morphism is nowhere dense, and base change of a finite morphism is finite. Morphisms, Lemmas morphisms-lemma-image-nowhere-dense-finite and morphisms-lemma-base-change-finite, invoked at proof lines 1686-1690. Supplies that the image of the intersection is not dense, so a rational point can be found outside it.

Source reference. Original lemma 038I: sources/stacks-varieties/pinned-original-source.tex, source lines 1542–1542. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04KX: sources/stacks-varieties/pinned-original-source.tex, source lines 1622–1622. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WL: sources/stacks-chow/tags.txt, source lines 2486–2486. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03HX: sources/stacks-chow/tags.txt, source lines 4583–4583. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. If X is irreducible and has a dense set of k-rational points, then X is geometrically irreducible.*

Proof. Let k'/k be a finite extension of fields and let $Z, Z' \subset X_{k'}$ be irreducible components. It suffices to show $Z = Z'$, see Lemma 038I. By Lemma 04KX we have $p(Z) = p(Z') = X$ where $p : X_{k'} \to X$ is the projection. If $Z \neq Z'$ then $Z \cap Z'$ is nowhere dense in $X_{k'}$ and hence $p(Z \cap Z')$ is not dense by Morphisms, Lemma 03HX; here we also use that p is a finite morphism as the base change of the finite morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$, see Morphisms, Lemma 01WL. Thus we can pick a k-rational point $x \in X$ with $x \notin p(Z \cap Z')$. Since the residue field of x is k we see that $p^{-1}(\{x\}) = \{x'\}$ where $x' \in X_{k'}$ is a point whose residue field is k'. Since $x \in p(Z) = p(Z')$ we conclude that $x' \in Z \cap Z'$ which is the contradiction we were looking for. $\square$